12

LAN system components

Objectives

When you have completed this chapter you should be able to:

- Explain the basic function of each of the devices listed under 12.1
- Explain the fundamental differences between the operation and application of switches (layer 2 and 3), bridges and routers

12.1 Introduction

In the design of an Ethernet system there are a number of different components that can be used. These include:

- Repeaters
- Media converters
- Bridges
- Hubs
- Switches
- Routers
- Gateways
- Print servers
- Terminal servers
- Remote access servers
- Time servers
- Thin servers

The lengths of LAN segments are limited due to physical and collision domain constraints and there is often a need to increase this range. This can be achieved by means of a number of interconnecting devices, ranging from repeaters to gateways. It may also be necessary to partition an existing network into separate networks for reasons of security or traffic overload.

In modern network devices the functions mentioned above are often mixed. Here are a few examples:

- A shared 10BaseT hub is, in fact, a multi-port repeater
- A Layer 2 switch is essentially a multi-port bridge
- Segmentable and dual-speed shared hubs make use of internal bridges
- Switches can function as bridges, a two-port switch being none other than a bridge
- Layer 3 switches function as routers

These examples are not meant to confuse the reader, but serve to emphasize the fact that the functions should be understood, rather than the 'boxes' in which they are packaged.

12.2 Repeaters

A repeater operates at the physical layer of the OSI model (layer 1) and simply retransmits incoming electrical signals. This involves amplifying and re-timing the signals received on one segment onto all other segments, without considering any possible collisions. All segments need to operate with the same media access mechanism and the repeater is unconcerned with the meaning of the individual bits in the packets. Collisions, truncated packets or electrical noise on one segment are transmitted onto all other segments.

12.2.1 Packaging

Repeaters are packaged either as stand-alone units (i.e. desktop models or small cigarette package-sized units) or 19" rack-mount units. Some of these can link two segments only, while larger rack-mount modular units (called concentrators) are used for linking multiple segments. Regardless of packaging, repeaters can be classified either as local repeaters (for linking network segments that are physically in close proximity), or as remote repeaters for linking segments that are some distance apart.

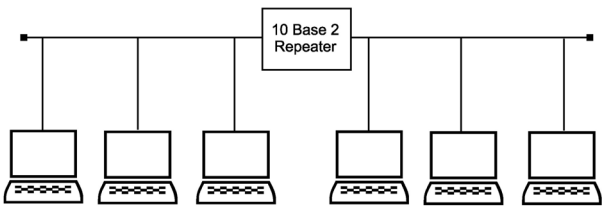


Figure 12.1
Repeater application

12.2.2 Local Ethernet repeaters

Several options are available:

- Two-port local repeaters offer most combinations of 10Base5, 10Base2, 10BaseT and 10BaseFL such as 10Base5/10Base5, 10Base2/10Base2, 10Base5/10Base2, 10Base2/10BaseT, 10BaseT/10BaseT and 10BaseFL/10BaseFL. By using such devices (often called boosters or extenders) one can, for example, extend the distance between a computer and

a 10BaseT Hub by up to 100m, or extend a 10BaseFL link between two devices (such as bridges) by up 2-3 km

- Multi-port local repeaters offer several ports of the same type (e.g. 4x 10Base2 or 8x 10Base5) in one unit, often with one additional connector of a different type (e.g. 10Base2 for a 10Base5 repeater.) In the case of 10BaseT the cheapest solution is to use an off-the-shelf 10BaseT shared hub, which is effectively a multi-port repeater
- Multi-port local repeaters are also available as chassis-type units; i.e. as frames with common back planes and removable units. An advantage of this approach is that 10Base2, 10Base5, 10BaseT and 10BaseFL can be mixed in one unit, with an option of SNMP management for the overall unit. These are also referred to as concentrators

12.2.3 Remote repeaters

Remote repeaters, on the other hand, have to be used in pairs with one repeater connected to each network segment and a fiber-optic link between the repeaters. On the network side they typically offer 10Base5, 10Base2 and 10BaseT. On the interconnecting side the choices include 'single pair Ethernet', using telephone cable up to 457m in length, or single mode/multi-mode optic fiber, with various connector options. With 10BaseFL (backwards compatible with the old FOIRL standard), this distance can be up to 1.6 km.

In conclusion it must be emphasized that although repeaters are probable the cheapest way to extend a network, they do so without separating the collision domains, or network traffic. They simply extend the physical size of the network. All segments joined by repeaters therefore share the same bandwidth and collision domain.

12.3 Media converters

Media converters are essentially repeaters, but interconnect mixed media viz. copper and fiber. An example would be 10BaseT/10BaseFL. As in the case of repeaters, they are available in single and multi-port options, and in stand-alone or chassis type configurations. The latter option often features remote management via SNMP.

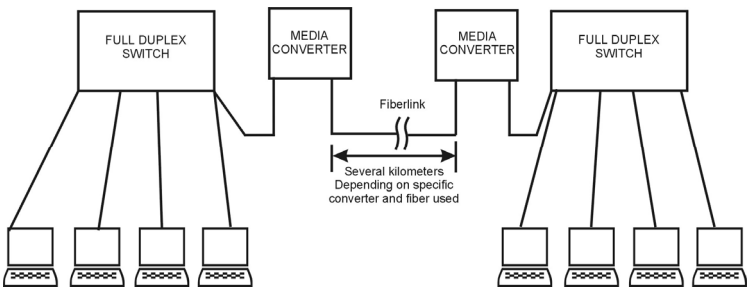


Figure 12.2
Media converter application

Models may vary between manufacturers, but generally Ethernet media converters support:

- 10 Mbps (10Base2, 10BaseT, 10BaseFL- single and multi-mode)
- 100 Mbps (Fast) Ethernet (100BaseTX, 100BaseFX- single and multimode)
- 1000 Mbps (Gigabit) Ethernet (single and multimode)

An added advantage of the fast and Gigabit Ethernet media converters is that they support full duplex operation that effectively doubles the available bandwidth.

12.4 Bridges

Bridges operate at the data link layer of the OSI model (layer 2) and are used to connect two separate networks to form a single large continuous LAN. The overall network, however, still remains one network with a single network ID (NetID). The bridge only divides the network up into two segments, each with its own collision domain and each retaining its full (say, 10 Mbps) bandwidth. Broadcast transmissions are seen by all nodes, on both sides of the bridge.

The bridge exists as a node on each network and passes only valid messages across to destination addresses on the other network. The decision as to whether or not a frame should be passed across the bridge is based on the layer 2 address, i.e. the media (MAC) address. The bridge stores the frame from one network and examines its destination MAC address to determine whether it should be forwarded across the bridge.

Bridges can be classified as either MAC or LLC bridges, the MAC sub-layer being the lower half of the data link layer and the LLC sub-layer being the upper half. For MAC bridges the media access control mechanism on both sides must be identical; thus it can bridge only Ethernet to Ethernet, token ring to token ring and so on. For LLC bridges, the data link protocol must be identical on both sides of the bridge (e.g. IEEE 802.2 LLC); however, the physical layers or MAC sub-layers do not necessarily have to be the same. Thus the bridge isolates the media access mechanisms of the networks. Data can therefore be transferred, for example, between Ethernet and token ring LANs. In this case, collisions on the Ethernet system do not cross the bridge nor do the tokens.

Bridges can be used to extend the length of a network (as with repeaters) but in addition they improve network performance. For example, if a network is demonstrating fairly slow response times, the nodes that mainly communicate with each other can be grouped together on one segment and the remaining nodes can be grouped together in another segment. The busy segment may not see much improvement in response rates (as it is already quite busy) but the lower activity segment may see quite an improvement in response times. Bridges should be designed so that 80% or more of the traffic is within the LAN and only 20% cross the bridge. Stations generating excessive traffic should be identified by a protocol analyzer and relocated to another LAN.

12.4.1 Intelligent bridges

Intelligent bridges (also referred to as transparent or spanning-tree bridges) are the most commonly used bridges because they are very efficient in operation and do not need to be taught the network topology. A transparent bridge learns and maintains two address lists corresponding to each network it is connected to. When a frame arrives from the one Ethernet network, its source address is added to the list of source addresses for that network. The destination address is then compared to that of the two lists of addresses for each network and a decision made whether to transmit the frame onto the other network. If no corresponding address to the destination node is recorded in either of these two lists the message is retransmitted to all other bridge outputs (flooding), to ensure the message is delivered to the correct network. Over a period of time, the bridge learns all the addresses on each network and thus avoids unnecessary traffic on the other network. The bridge also maintains time out data for each entry to ensure the table is kept up to date and old entries purged.

Transparent bridges cannot have loops that could cause endless circulation of packets. If the network contains bridges that could form a loop as shown in Figure 12.3, one of the bridges (C) needs to be made redundant and deactivated.

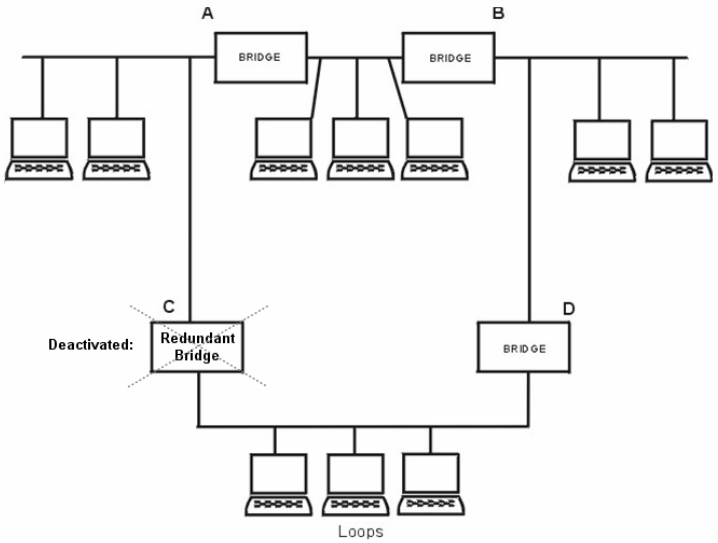


Figure 12.3
Avoidance of loops in bridge networks

The spanning tree algorithm (IEEE 802.1d) is used to manage paths between segments having redundant bridges. This algorithm designates one bridge in the spanning tree as the root and all other bridges transmit frames towards the root using a least cost metric. Redundant bridges can be reactivated if the network topology changes.

12.4.2 Source-routing bridges

Source routing (SR) Bridges are popular for IBM token ring networks. In these networks, the sender must determine the best path to the destination. This is done by sending a discovery frame that circulates the network and arrives at the destination with a record of the path token. These frames are returned to the sender who can then select the best path. Once the path has been discovered, the source updates its routing table and includes the path details in the routing information field in the transmitted frame.

12.4.3 SRT and translational bridges

When connecting Ethernet networks to token ring networks, either **source routing transparent** (SRT) bridges or translational bridges are used. SRT bridges are a combination of a transparent and source routing bridge, and are used to interconnect Ethernet (IEEE802.3) and token ring (IEE802.5) networks. It uses source routing of the data frame if it contains routing information; otherwise it reverts to transparent bridging. Translational bridges, on the other hand, translate the routing information to allow source routing networks to bridge to transparent networks. The IBM 8209 is an example of this type of bridge.

12.4.4 Local vs. remote bridges

Local bridges are devices that have two network ports and hence interconnect two adjacent networks at one point. This function is currently often performed by switches, being essentially intelligent multi-port bridges.

A very useful type of local bridge is a 10/100 Mbps Ethernet bridge, which allows interconnection of 10BaseT, 100BaseTX and 100BaseFX networks, thereby performing the required speed translation. These bridges typically provide full duplex operation on 100BastTX and 100BaseFX, and employ internal buffers to prevent saturation of the 10BaseT port.

Remote bridges, on the other hand, operate in pairs with some form of interconnection between them. This interconnection can be with or without modems, and include

RS-232/V.24, V.35, RS-422, RS-530, X.21, 4-wire, or fiber (both single and multi-mode). The distance between bridges can typically be up to 1.6 Km.

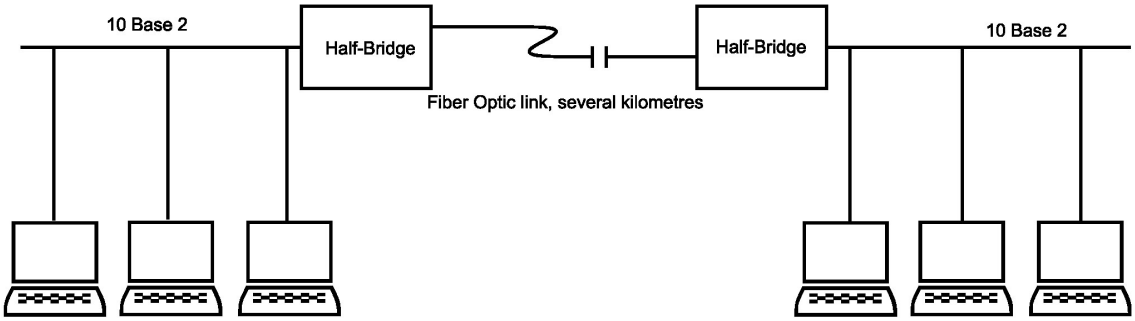


Figure 12.4
Remote bridge application

12.5 Hubs

Hubs are used to interconnect hosts in a physical star configuration. This section will deal with Ethernet hubs, which are of the 10/100/100BaseT variety. They are available in many configurations, some of which will be discussed below.

12.5.1 Desktop vs stackable hubs

Smaller desktop units are intended for stand-alone applications, and typically have 5 to 8 ports. Some 10BaseT desktop models have an additional 10Base2 port. These devices are often called workgroup hubs.

Stackable hubs, on the other hand, typically have up to 24 ports and can be physically stacked and interconnected to act as one large hub without any repeater count restrictions. These stacks are often mounted in 19-inch cabinets.

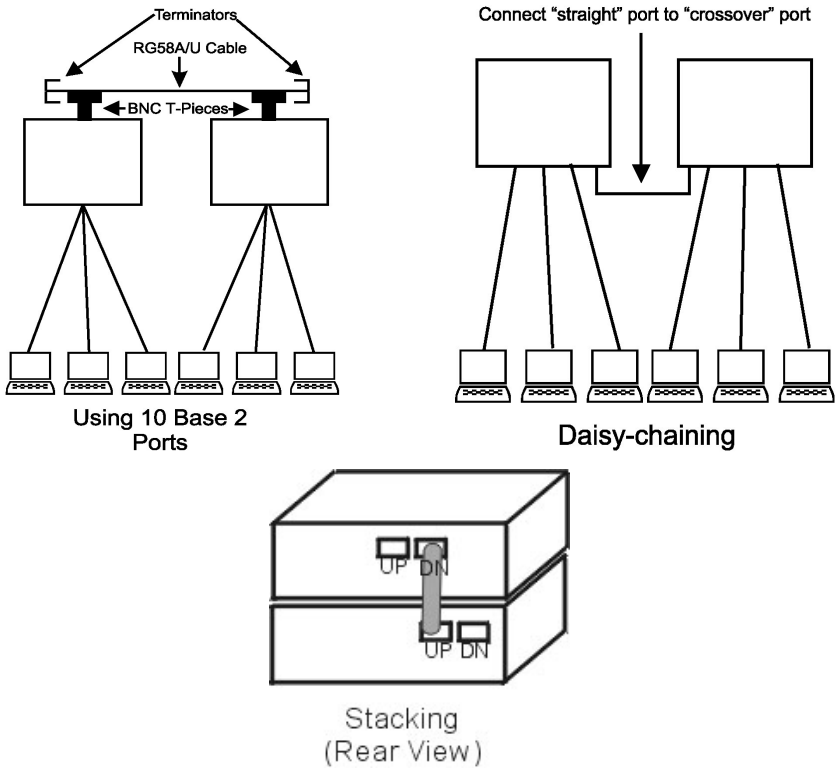


Figure 12.5
10BaseT hub interconnection

12.5.2 Shared vs. switched hubs

Shared hubs interconnect all ports on the hub in order to form a logical bus. This is typical of the cheaper workgroup hubs. All hosts connected to the hub share the available bandwidth since they all form part of the same collision domain.

Although they physically look alike, switched hubs (better known as switches) allow each port to retain and share its full bandwidth only with the hosts connected to that port. Each port (and the segment connected to that port) functions as a separate collision domain. This attribute will be discussed in more detail in the section on switches.

12.5.3 Managed hubs

Managed hubs have an on-board processor with its own MAC and IP address. Once the hub has been set up via a PC on the hub's serial (COM) port, it can be monitored and controlled via the network using SNMP or Telnet. The user can perform activities such as enabling/ disabling individual ports, performing segmentation (see next section), monitoring the traffic on a given port, or setting alarm conditions for a given port.

12.5.4 Segmentable hubs

On a non-segmentable (i.e. shared) hub, all hosts share the same bandwidth. On a segmentable hub, however, the ports can be grouped, under software control, into several shared groups. All hosts on each segment then share the full bandwidth on that segment, which means that a 24-port 10BaseT hub segmented into 4 groups effectively supports 40

Mbps. The configured segments are internally connected via bridges, so that all ports can still communicate with each other if needed.

12.5.5 Dual-speed hubs

Some hubs offer dual-speed ports, e.g. 10BaseT/100BaseT. These ports are auto-configured, i.e. each port senses the speed of the NIC connected to it, and adjusts its own speed accordingly. All the 10BaseT ports connect to a common low-speed internal segment, while all the 100BaseT ports connect to a common high-speed internal segment. The two internal segments are interconnected via a speed-matching bridge.

12.5.6 Modular hubs

Some stackable hubs are modular, allowing the user to configure the hub by plugging in a separate module for each port. Ethernet options typically include both 10 and 100 Mbps, with either copper or fiber. These hubs are sometimes referred to as chassis hubs.

12.5.7 Hub interconnection

Stackable hubs are best interconnected by means of special stacking cables attached to the appropriate connectors on the back of the chassis.

An alternative method for non-stackable hubs is by 'daisy-chaining' an interconnecting port on each hub by means of a UTP patch cord. Care has to be taken not to connect the 'transmit' pins on the ports together (and, for that matter, the 'receive' pins) – it simply will not work. This is similar to interconnecting two COM ports with a 'straight' cable i.e. without a null modem. Connect transmit to receive and vice versa by (a) using a crossover cable and interconnecting two 'normal' ports, or (b) using a normal ('straight') cable and utilizing a crossover port on one of the hubs. Some hubs have a dedicated uplink (crossover) port while others have a port that can be manually switched into crossover mode.

A third method that can be used on hubs with a 10Base2 port is to create a backbone. Attach a BNC T-piece to each hub, and interconnect the T-pieces with RG-58 coax cable. The open connections on the extreme ends of the backbone obviously have to be terminated.

Fast Ethernet hubs need to be deployed with caution because the inherent propagation delay of the hub is significant in terms of the 5.12 microsecond collision domain size. Fast Ethernet hubs are classified as Class I, II or II+, and the class dictates the number of hubs that can be interconnected. For example, Class II dictates that there may be no more than two hubs between any given pair of nodes, that the maximum distance between the two hubs shall not exceed 5 m, and that the maximum distance between any two nodes shall not exceed 205 m. The safest approach, however, is to follow the guidelines of each manufacturer.

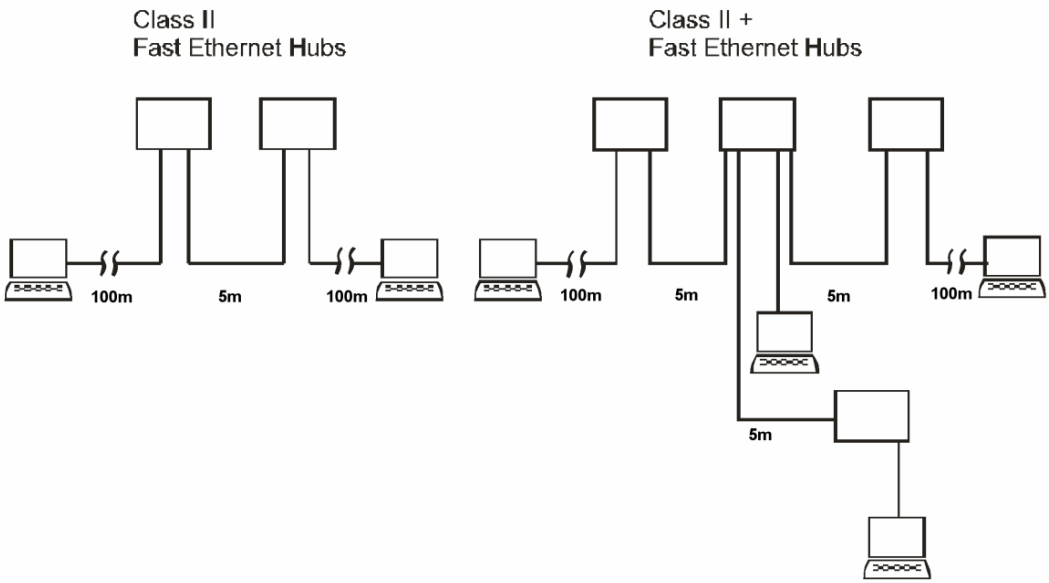


Figure 12.6
Fast Ethernet hub interconnection

12.6 Switches

Ethernet switches are an expansion of the concept of bridging and are, in fact, intelligent (self-learning) multi-port bridges. They enable frame transfers to be accomplished between any pair of devices on a network, on a per-frame basis. Only the two ports involved 'see' the specific frame. Illustrated below is an example of an 8 port switch, with 8 hosts attached. This comprises a physical star configuration, but it does not operate as a logical bus as an ordinary hub does. Since each port on the switch represents a separate segment with its own collision domain, it means that there are only 2 devices on each segment, namely the host and the switch port. Hence, in this particular case, there can be no collisions on any segment!

In the sketch below hosts 1 & 7, 3 & 5 and 4 & 8 need to communicate at a given moment, and are connected directly for the duration of the frame transfer. For example, host 7 sends a packet to the switch, which determines the destination address, and directs the package to port 1 at 10 Mbps.

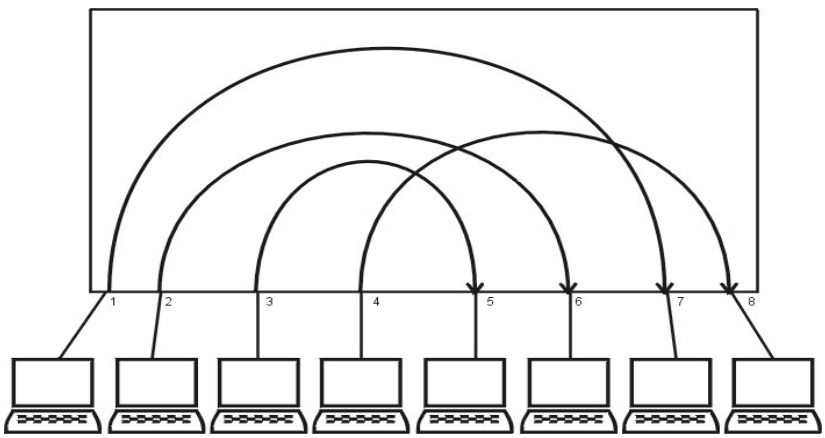


Figure 12.7
8-Port Ethernet switch

If host 3 wishes to communicate with host 5, the same procedure is repeated. Provided that there are no conflicting destinations, a 16-port switch could allow 8 concurrent frame exchanges at 10 Mbps, rendering an effective bandwidth of 80 Mbps. On top of this, the switch could allow full duplex operation, which would double this figure.

12.6.1 Cut-through vs store-and-forward

Switches have two basic architectures, cut-through and store-and-forward. In the past, cut-through switches were faster because they examined the packet destination address only before forwarding the frame to the destination segment. A store-and-forward switch, on the other hand, accepts and analyses the entire packet before forwarding it to its destination. It takes more time to examine the entire packet, but it allows the switch to catch certain packet errors and keep them from propagating through the network. The speed of modern store-and-forward switches has caught up with cut-through switches so that the speed difference between the two is minimal. There are also a number of hybrid designs that mix the two architectures.

Since a store-and-forward switch buffers the frame, it can delay forwarding the frame if there is traffic on the destination segment, thereby adhering to the CSMA/CD protocol. In the case of a cut-through switch this is a problem, since a busy destination segment means that the frame cannot be forwarded, yet it cannot be stored either. The solution is to force a collision on the source segment, thereby enticing the source host to retransmit the frame.

12.6.2 Layer 2 switches vs. layer 3 switches

Layer 2 switches operate at the data link layer of the OSI model and derive their addressing information from the destination MAC address in the Ethernet header. Layer 3 switches, on the other hand, obtain addressing information from the Network Layer, namely from the destination IP address in the IP header. Layer 3 switches are used to replace routers in LANs as they can do basic IP routing (supporting protocols such as RIP and RIPv2) at almost ‘wire-speed’; hence they are significantly faster than routers.

12.6.3 Full duplex switches

An additional advancement is full duplex Ethernet where a device can simultaneously transmit AND receive data over one Ethernet connection. This requires a different Ethernet NIC in the host, as well as a switch that supports full duplex. This enables two devices two transmit and receive simultaneously via a switch. The node automatically negotiates with the switch and uses full duplex if both devices can support it.

Full duplex is useful in situations where large amounts of data are to be moved around quickly, for example between graphics workstations and file servers.

12.6.4 Switch applications

High-speed aggregation

Switches are very efficient in providing a high-speed aggregated connection to a server or backbone. Apart from the normal lower-speed (say, 10BaseT) ports, switches have a high-speed uplink port (100BaseTX). This port is simply another port on the switch, accessible by all the other ports, but features a speed conversion from 10 Mbps to 100 Mbps.

Assume that the uplink port was connected to a file server. If all the other ports (say, eight times 10BaseT) wanted to access the server concurrently, this would necessitate a bandwidth of 80 Mbps in order to avoid a bottleneck and subsequent delays. With a 10BaseT uplink port this would create a serious problem. However, with a 100BaseTX uplink there is still 20 Mbps of bandwidth to spare.

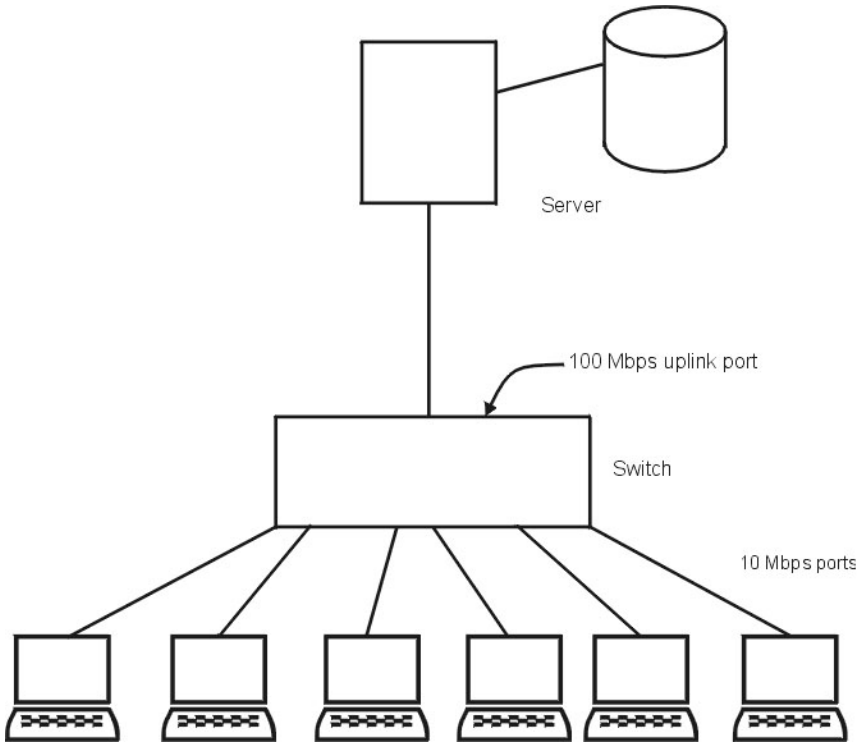


Figure 12.8
Using a Switch to connect users to a Server

Backbones

Switches are very effective in backbone applications, linking several LANs together as one, yet segregating the collision domains. An example could be a switch located in the basement of a building, linking the networks on different floors of the building. Since the actual 'backbone' is contained within the switch, it is known in this application as a 'collapsed backbone'.

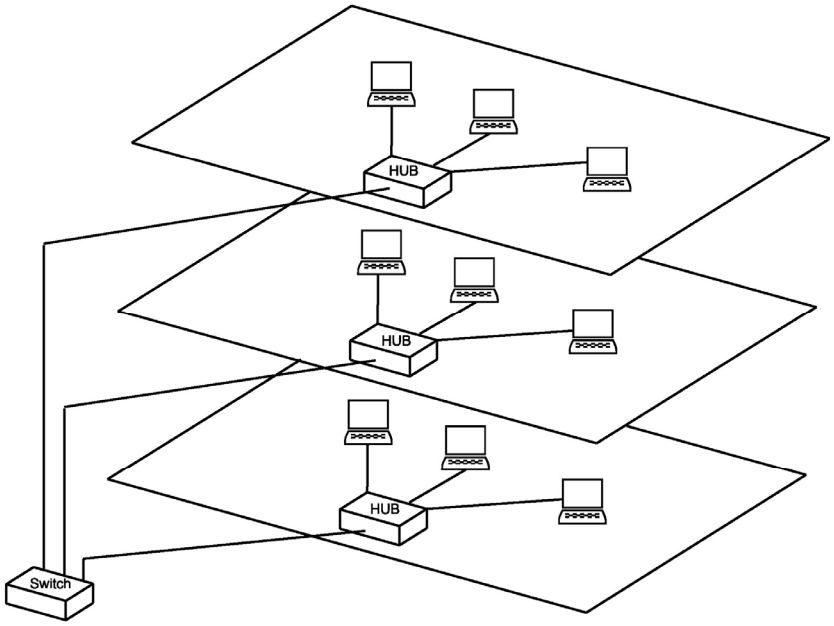


Figure 12.9
Using a switch as a backbone

VLANs and deterministic Ethernet

Provided that a LAN is constructed around switches that support VLANs, individual hosts on the physical LAN can be grouped into smaller Virtual LANs (VLANs), totally invisible to their fellow hosts. Unfortunately, neither the 'standard' Ethernet nor the IEEE802.3 header contains sufficient information to identify members of each VLAN; hence, the frame had to be modified by the insertion of a 'tag', between the Source MAC address and the type/length fields. This modified frame is known as an Ethernet 802.1Q tagged frame and is used for communication between the switches.

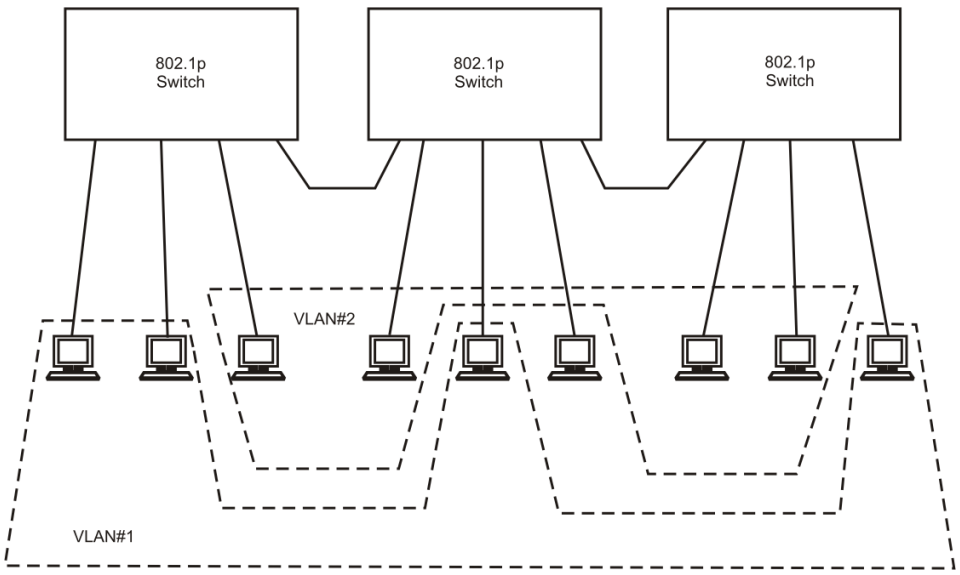


Figure 12.10
Virtual LAN's using switches

The IEEE802.1p committee has defined a standard for packet-based LAN's that supports Layer 2 traffic prioritization in a switched LAN environment. IEEE802.1p is

part of a larger initiative (IEEE802.1p/Q) that adds more information to the Ethernet header (as shown in Fig 12.11) to allow networks to support VLANs and traffic prioritization.

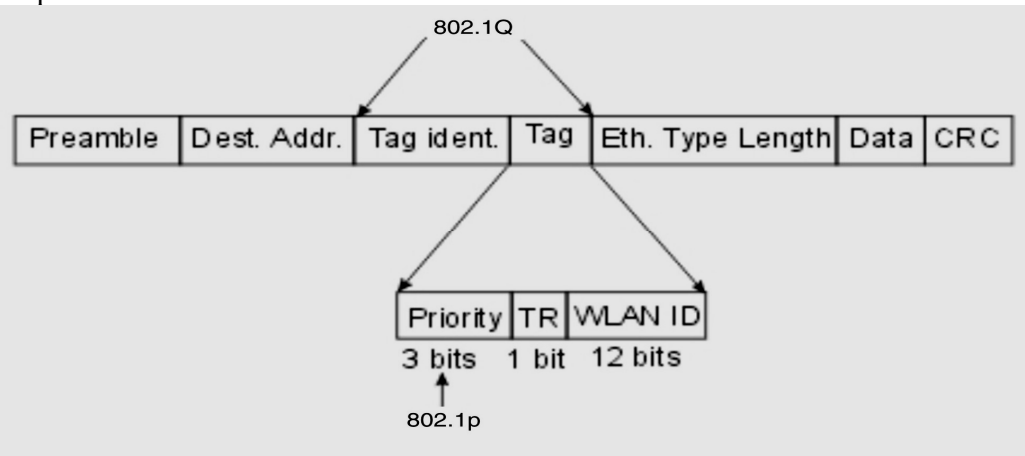


Figure 12.11
IEEE802.1p/Q modified Ethernet header

802.1p/Q adds 16 bits to the header, of which three are for a priority tag and twelve for a VLAN ID number. This allows for eight discrete priority Layers from 0 (high) to 7 (low) that supports different kinds of traffic in terms of their delay-sensitivity. Since IEEE802.1p/Q operates at Layer II, it supports prioritization for all traffic on the VLAN, both IP and non-IP. This introduction of priority layers enables so-called deterministic Ethernet where, instead of contending for access to a bus, a source node can pass a frame directly to a destination node on the basis of its priority, and without risk of any collisions.

The TR bit simply indicates whether the MSB of the VLAN ID is on the left or the right.

12.7 Routers

Unlike bridges and layer 2 switches, routers operate at layer 3 of the OSI model, namely at the network layer (or, the Internet layer of the DOD model). They therefore ignore address information contained within the data link layer (the MAC addresses) and rather delve deeper into each frame and extract the address information contained in the network layer. For TCP/IP this is the IP address.

Like bridges or switches, routers appear as hosts on each network that it is connected to. They are connected to each participating network through an NIC, each with an MAC address as well as an IP address. Each NIC has to be assigned an IP address with the same NetID as the network it is connected to. This IP address allocated to each network is known as the default gateway for that network and each host on the internetwork requires at least one default gateway (but could have more). The default gateway is the IP address to which any host must forward a packet if it finds that the NetID of the destination and the local NetID does not match, which implies remote delivery of the packet.

A second major difference between routers and bridges or switches is that routers will not act autonomously but rather have to be GIVEN the frames that need to be forwarded. A host to the designated Default Gateway forwards such frames.

Protocol dependency

Because routers operate at the network layer, they are used to transfer data between two networks that have the same Internet layer protocols (such as IP) but not necessarily the same physical or data link protocols. Routers are therefore said to be protocol dependent, and have to be able to handle all the Internet layer protocols present on a particular network. A network utilizing Novell Netware therefore requires routers that can accommodate IPX (Internet packet exchange) – the network layer component of SPX/IPX. If this network has to handle Internet access as well, it can only do this via IP, and hence the routers will need to be upgraded to models that can handle both IPX and IP.

Routers maintain tables of the networks that they are connected to and of the optimum path to reach a particular network. They then redirect the message to the next router along that path.

12.7.1 Two-port vs. multi-port routers

Multi-port routers are chassis-based devices with modular construction. They can interconnect several networks. The most common type of router is, however, a 2-port router. Since these are invariably used to implement WANs, they connect LANs to a 'communications cloud'; the one port will be a local LAN port e.g. 10BaseT, but the second port will be a WAN port such as X.25.

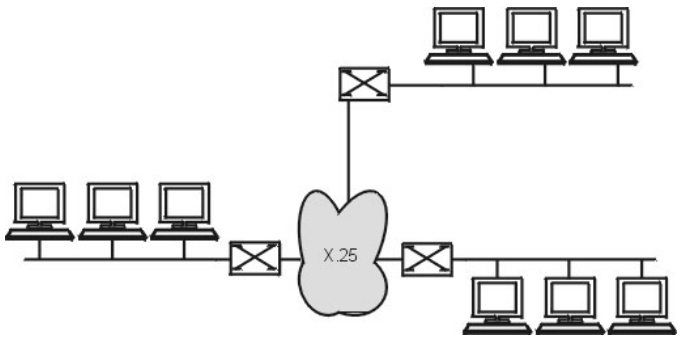


Figure 12.12
Implementing a WAN with 2-port routers (gateways)

12.7.2 Access routers

Access routers are 2-port routers that use dial-up access rather than a permanent (e.g. X.25) connection to connect a LAN to an ISP and hence to the 'communications cloud' of the Internet. Typical options are ISDN or dial-up over telephone lines, using either the V.34 (ITU 33.6kbps) or V.90 (ITU 56kbps) standard. Some models allow multiple phone lines to be used, using multi-link PPP, and will automatically dial up a line when needed or redial when a line is dropped, thereby creating a 'virtual leased line'.

12.7.3 Border routers

Routers within an autonomous system normally communicate with each other using an interior gateway protocol such as RIP. However, routers within an autonomous system that also communicate with remote autonomous systems need to do that via an exterior gateway protocol such as BGP-4. Whilst doing this, they still have to communicate with

other routers within their own autonomous system, e.g. via RIP. These routers are referred to as border routers.

12.7.4 Routing vs. bridging

It sometimes happens that a router is confronted with a layer 3 (network layer) address it does not understand. In the case of an IP router, this may be a Novell IPX address. A similar situation will arise in the case of NetBIOS/NetBEUI, which is non-routable. A 'brouter' (bridging router) will revert to a bridge and try to deal with the situation at layer 2 if it cannot understand the layer 3 protocol, and in this way forward the packet towards its destination. Most modern routers have this function built in.

12.8 Gateways

Gateways are network interconnection devices, not to be confused with 'default gateways' which are the ROUTER IP addresses to which packets are forwarded for subsequent routing (indirect delivery).

A gateway is designed to connect dissimilar networks and could operate anywhere from layer 4 to layer 7 of the OSI model. In a worst case scenario, a gateway may be required to decode and re-encode all seven layers of two dissimilar networks connected to either side, for example when connecting an Ethernet network to an IBM SNA network. Gateways thus have the highest overhead and the lowest performance of all the internetworking devices. The gateway translates from one protocol to the other and handles differences in physical signals, data format, and speed.

Since gateways are, per definition, protocol converters, it so happens that a 2-port (WAN) router could also be classified as a 'gateway' since it has to convert both layer 1 and layer 2 on the LAN side (say, Ethernet) to layer 1 and Layer 2 on the WAN side (say, X.25). This leads to the confusing practice of referring to (WAN) routers as gateways.

12.9 Print servers

Print servers are devices, attached to the network, through which printers can be made available to all users. Typical print servers cater for both serial and parallel printers. Some also provide concurrent multi-protocol support, which means that they support multiple protocols and will execute print jobs on a first-come first-served basis regardless of the protocol used. Protocols supported could include SPX/IPX, TCP/IP, AppleTalk/EtherTalk, NetBIOS/NetBEUI, or DEC LAT.

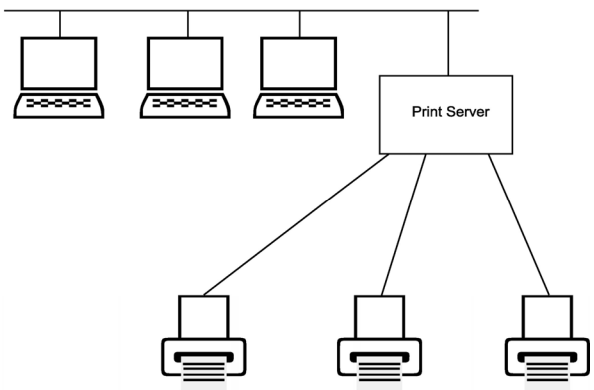


Figure 12.13
Print server applications

12.10 Terminal servers

Terminal servers connect multiple (typically up to 32) serial (RS-232) devices such as system consoles, data entry terminals, bar code readers, scanners, and serial printers to a network. They support multiple protocols such as TCP/IP, SPX/IPX, NetBIOS/NetBEUI, AppleTalk and DEC LAT, which means that they not only can handle devices which support different protocols, but that they can also provide protocol translation between ports.

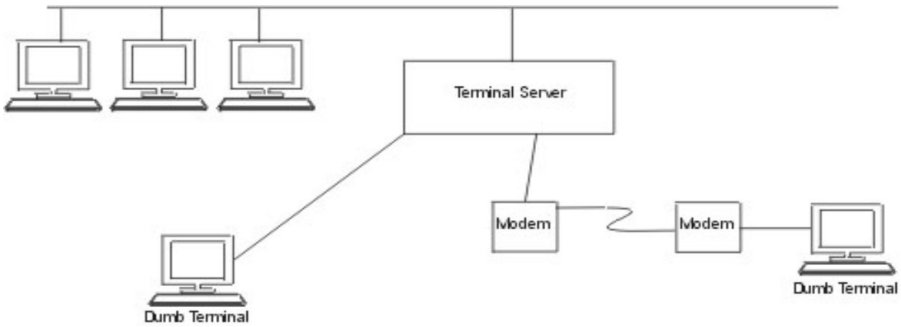


Figure 12.14
Terminal server applications

12.11 Thin servers

Thin servers are essentially single-channel terminal servers. They provide connectivity between Ethernet (10BaseT/100BaseTX) and any serial devices with RS-232 or RS-485 ports. They implement the bottom 4 layers of the OSI model with Ethernet and layer 3/4 protocols such as TCP/IP, SPX/IPX and DEC LAT.

A special version, the industrial thin server, is mounted in a rugged DIN rail package. It can be configured over one of its serial ports, and managed via Telnet or SNMP. A software redirector package enables a user to remove a serial device such as a weigh-bridge from its controlling computer, locate it elsewhere, then connect it via a thin server to an Ethernet network through the nearest available hub. All this is done without modifying any software. A software package called a port redirector makes the computer 'think' that it is still communicating via the weighbridge via the COM port while, in fact, the data and control messages to the device are routed via the network.

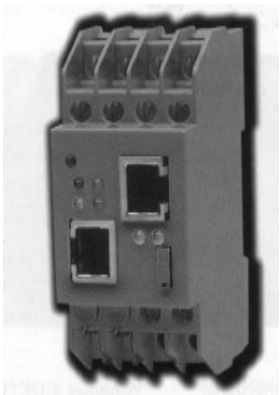


Figure 12.15
Industrial thin server (courtesy of Lantronix)

12.12 Remote access servers

Remote access servers are devices that allow users to dial into a network via analog telephone or ISDN. Typical remote access servers support between 1 and 32 dial-in users via PPP or SLIP. User authentication can be done via Radius, Kerberos or SecurID. Some offer dial-back facilities whereby the user authenticates to the server's internal table, after which the server dials back the user so that the cost of the connection is carried by the network and not the remote user.

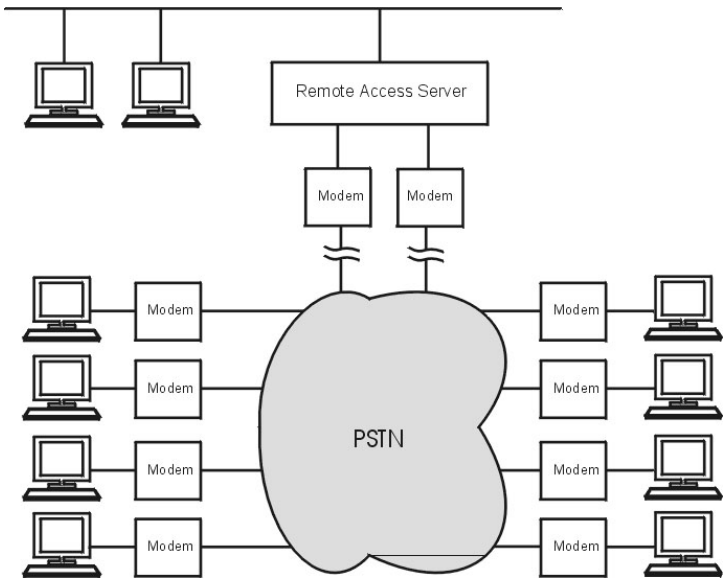


Figure 12.16
Remote access server application

12.13 Network timeservers

Network time servers are standalone devices that compute the correct local time my means of a global positioning system (GPS) receiver, and then distribute it across the network by means of the network time protocol (NTP).

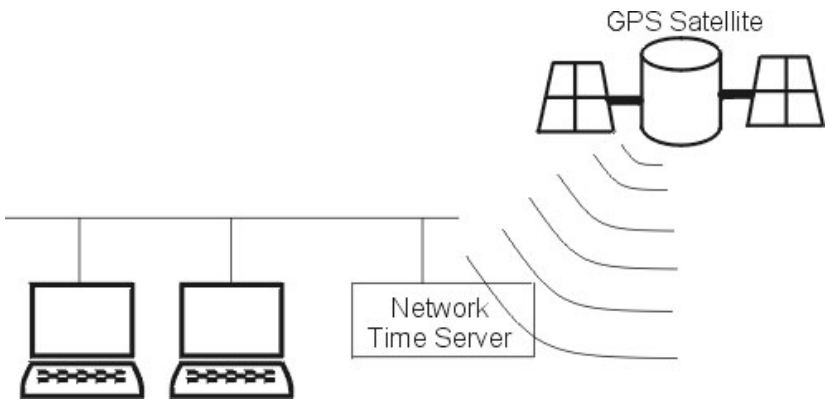


Figure 12.17
Network timeserver application